



CAUSAL MAPPING APPROACHES DIFFER IN APPLICATION, CONSTRUCTION, ANALYSIS AND HOW THEY DEAL WITH MULTIPLE SOURCES

Reference	Main application of causal mapping	Mode of construction	Dealing with multiple sources	Analysis procedures
(Axelrod 1976)	Understand and critique decision making	Coding documents	Mainly idiographic	Compute polarity of indirect effects in some cases.
(Bougon et al. 1977)	Understand how organisations are constructed and can be influenced.	Semi-structured interview to identify a fixed list of factors aka 'variables'; respondents then say which are linked and give the polarity.	Compare individual maps and combine into global 'average' map.	Identify variables X with high outdegree and Y with high indegree and construct an 'etiograph' to show all the multiple paths from one point to another; discuss how respondents might have influence over some variables.
(Ackermann et al. 2004; Ackermann & Eden 2011; Eden 1992; Eden et al. 1979; Eden et al. 1992)	Decision support and problem solving in organisations. Maps are seen primarily as useful tools rather than research about reality.	Open interviewing of several respondents based on Kelly's Personal Construct Theory. Also map construction directly with groups (1988).	Comparing maps between individuals and analysing group maps directly.	Various structural measures, presence of isolated clusters, hierarchical trees, loops. Simplify individual maps by collapsing X->Y->Z into X->Z.
(Laukkanen 1994; Laukkanen 2012; Laukkanen & Eriksson 2013; Laukkanen & Wang 2016)	Explicitly cognitive, to improve knowledge and understanding in management	Systematic comparative method with semi-structured interviewing: respondents are given anchor topic(s) then asked for causes,	Comparative study of different individual maps, combining data into a database.	Display combined maps for subgroups, e.g. all local managers.

Reference	Main application of causal mapping	Mode of construction	Dealing with multiple sources	Analysis procedures
		effects, causes of causes, effects of effects. Compress the data by standardising factor names. Comprehensive coverage of different map construction possibilities.		

References

- Ackermann, Eden, & Cropper (2004). *Getting Started with Cognitive Mapping*.
- Ackermann, & Eden (2011). *Using Causal Mapping to Support Information Systems Development*. Causal Mapping for Research in Information Technology.
- Axelrod (1976). *Structure of Decision: The Cognitive Maps of Political Elites*. Princeton university press.
- Bougon, Weick, & Binkhorst (1977). *Cognition in Organizations: An Analysis of the Utrecht Jazz Orchestra*. Administrative Science Quarterly, 606--639.
- Eden, Jones, & Sims (1979). *Thinking in Organisations*. Macmillan.
- Eden (1992). *On the Nature of Cognitive Maps*. Journal of Management Studies, 29, 261--265.
- Eden, Ackermann, & Cropper (1992). *The Analysis of Cause Maps*. Journal of Management Studies, 29, 309--324.
- Laukkanen (1994). *Comparative Cause Mapping of Organizational Cognitions*. Organization Science, 5, 322--343.
- Laukkanen (2012). *Comparative Causal Mapping and CMAP3 Software in Qualitative Studies*. Forum Qualitative Sozialforschung / Forum: Qualitative Social Research, 13. <https://doi.org/10.17169/fqs-13.2.1846>.
- Laukkanen, & Eriksson (2013). *New Designs and Software for Cognitive Causal Mapping*. Qualitative Research in Organizations and Management: An International Journal, 8, 122--147.
- Laukkanen, & Wang (2016). *Comparative Causal Mapping: The CMAP3 Method*. Routledge.